

In [ ]:

```
import pandas as pd
df=pd.read_csv('/content/agriculture.csv')
print(df)
```

|      | rainfall_mm | temperature_c | fertilizer_kg | irrigation_hours | soil_ph | \ |
|------|-------------|---------------|---------------|------------------|---------|---|
| 0    | 588.6       | 18.6          | 242.4         | 6.5              | 6.5     |   |
| 1    | 772.8       | 34.6          | 247.2         | 10.0             | 6.5     |   |
| 2    | 970.9       | 36.3          | 168.4         | 7.3              | 6.4     |   |
| 3    | 611.7       | 19.0          | 121.7         | 3.7              | 6.0     |   |
| 4    | 696.1       | 29.6          | 184.6         | 5.1              | 6.1     |   |
| ...  | ...         | ...           | ...           | ...              | ...     |   |
| 1495 | 473.8       | 32.3          | 65.0          | 3.6              | 5.8     |   |
| 1496 | 825.9       | 18.4          | 200.8         | 9.9              | 6.7     |   |
| 1497 | 329.3       | 23.3          | 90.3          | 7.7              | 7.0     |   |
| 1498 | 801.7       | 28.6          | 204.1         | 3.4              | 6.9     |   |
| 1499 | 684.8       | 23.7          | 88.9          | 1.2              | 5.9     |   |

|      | crop_type | soil_type | yield_ton_per_hectare |
|------|-----------|-----------|-----------------------|
| 0    | Maize     | Loamy     | 5.92                  |
| 1    | Maize     | Sandy     | 6.24                  |
| 2    | Wheat     | Sandy     | 4.77                  |
| 3    | Maize     | Loamy     | 4.04                  |
| 4    | Cotton    | Clay      | 5.17                  |
| ...  | ...       | ...       | ...                   |
| 1495 | Cotton    | Sandy     | 3.28                  |
| 1496 | Cotton    | Sandy     | 5.16                  |
| 1497 | Soybean   | Sandy     | 3.60                  |
| 1498 | Rice      | Sandy     | 5.03                  |
| 1499 | Maize     | Clay      | 3.42                  |

[1500 rows x 8 columns]

In [10]:

```
print(f"Rows and Columns: {df.shape}")
print(f"Columns: {list(df.columns)}")
print("\nFirst 10 Records:")
print(df.head(10))
```

Rows and Columns: (1500, 8)

Columns: ['rainfall\_mm', 'temperature\_c', 'fertilizer\_kg', 'irrigation\_hours', 'soil\_ph', 'crop\_type', 'soil\_type', 'yield\_ton\_per\_hectare']

First 10 Records:

|   | rainfall_mm | temperature_c | fertilizer_kg | irrigation_hours | soil_ph | \ |
|---|-------------|---------------|---------------|------------------|---------|---|
| 0 | 588.6       | 18.6          | 242.4         | 6.5              | 6.5     |   |
| 1 | 772.8       | 34.6          | 247.2         | 10.0             | 6.5     |   |
| 2 | 970.9       | 36.3          | 168.4         | 7.3              | 6.4     |   |
| 3 | 611.7       | 19.0          | 121.7         | 3.7              | 6.0     |   |
| 4 | 696.1       | 29.6          | 184.6         | 5.1              | 6.1     |   |
| 5 | 831.9       | 28.0          | 190.3         | 2.1              | 6.1     |   |
| 6 | 1023.8      | 32.0          | 108.5         | 6.9              | 6.2     |   |
| 7 | 1142.4      | 18.4          | 241.9         | 4.1              | 7.3     |   |
| 8 | 810.4       | 36.4          | 164.8         | 9.9              | 6.5     |   |
| 9 | 1085.5      | 29.4          | 89.3          | 8.3              | 5.8     |   |

|   | crop_type | soil_type | yield_ton_per_hectare |
|---|-----------|-----------|-----------------------|
| 0 | Maize     | Loamy     | 5.92                  |
| 1 | Maize     | Sandy     | 6.24                  |
| 2 | Wheat     | Sandy     | 4.77                  |
| 3 | Maize     | Loamy     | 4.04                  |
| 4 | Cotton    | Clay      | 5.17                  |
| 5 | Soybean   | Loamy     | 5.50                  |
| 6 | Cotton    | Loamy     | 5.44                  |
| 7 | Soybean   | Sandy     | 5.88                  |
| 8 | Wheat     | Loamy     | 6.35                  |
| 9 | Cotton    | Loamy     | 6.30                  |

In [11]:

```
print("\nData Types:")
print(df.dtypes)
print(f"\nMissing values per column:\n{df.isnull().sum()}")
```

Data Types:  
rainfall\_mm float64  
temperature\_c float64  
fertilizer\_kg float64  
irrigation\_hours float64  
soil\_ph float64  
crop\_type object  
soil\_type object  
yield\_ton\_per\_hectare float64  
dtype: object

Missing values per column:  
rainfall\_mm 0  
temperature\_c 0  
fertilizer\_kg 0  
irrigation\_hours 0  
soil\_ph 0  
crop\_type 0  
soil\_type 0  
yield\_ton\_per\_hectare 0  
dtype: int64

In [12]:

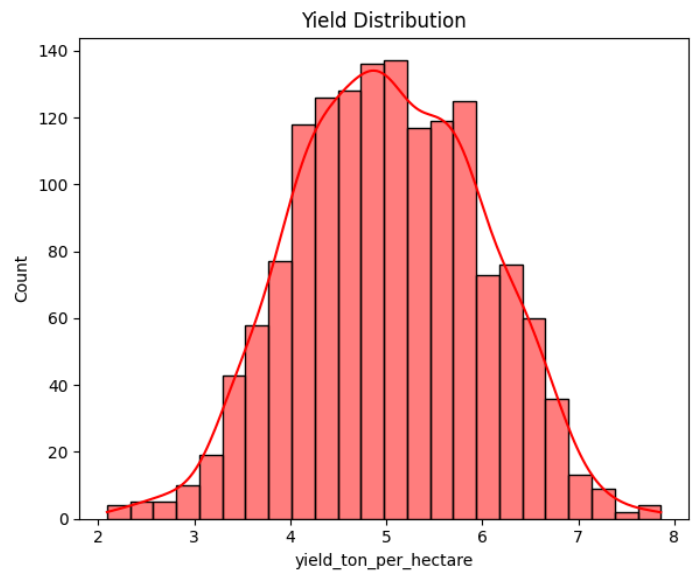
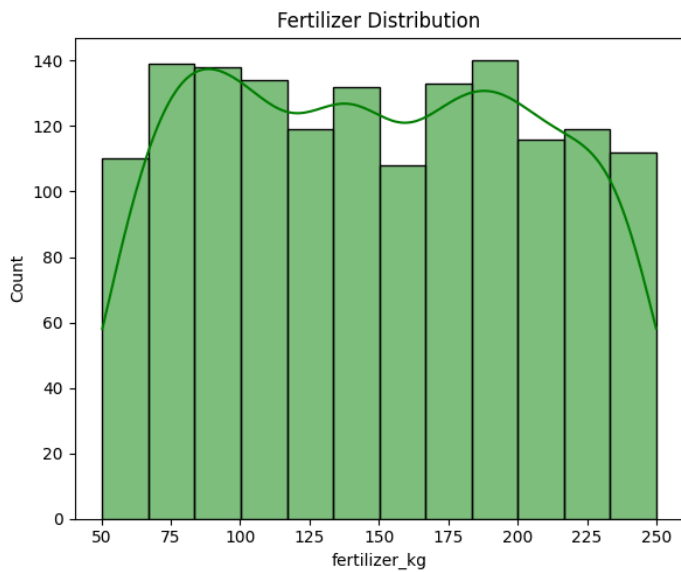
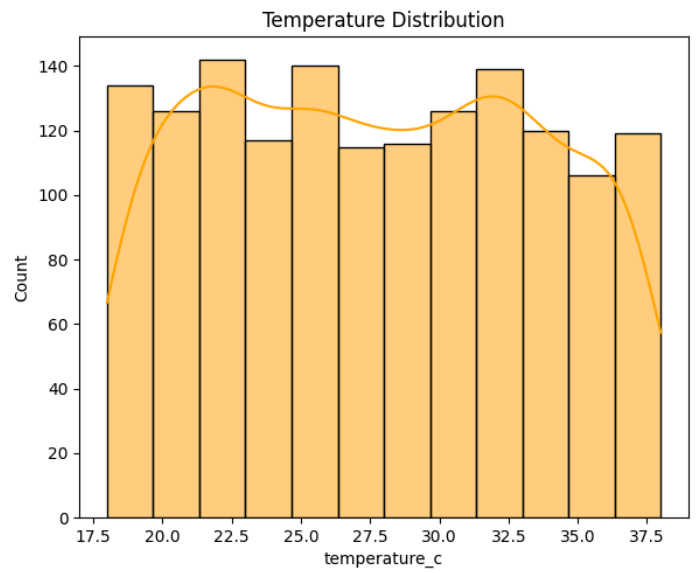
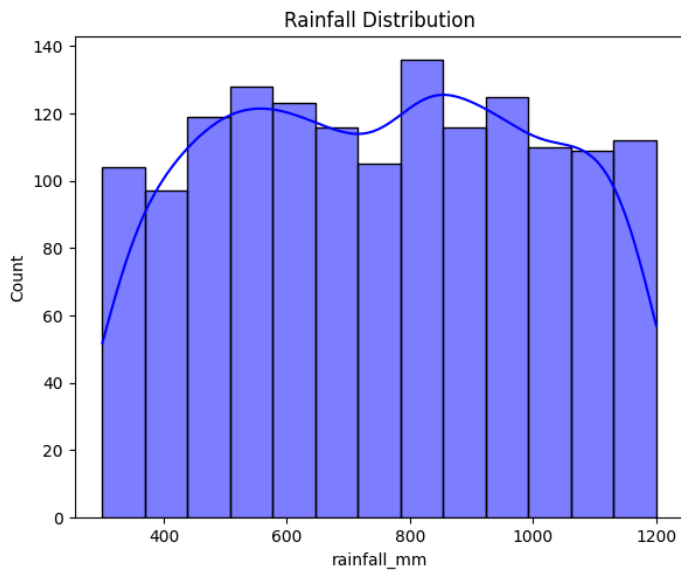
```
stats = df.describe()
print("\nSummary Statistics:")
print(stats)
print(f"Highest Mean: {stats.loc['mean'].idxmax()}")
print(f"Highest Std Dev: {stats.loc['std'].idxmax()}")
```

```
Summary Statistics:
count    rainfall_mm  temperature_c  fertilizer_kg  irrigation_hours  \
count    1500.000000    1500.000000    1500.000000    1500.000000
mean      754.054667      27.749467    148.744067      5.403267
std       255.097216      5.758101     56.990279      2.584329
min       300.200000     18.000000     50.300000      1.000000
25%       536.175000     22.600000     98.600000      3.200000
50%       761.200000     27.700000    146.850000      5.400000
75%       964.375000     32.600000    196.575000      7.600000
max      1200.000000     38.000000    249.900000     10.000000

count    soil_ph  yield_ton_per_hectare
count    1500.000000    1500.000000
mean      6.759133      5.028793
std       0.719742      0.968282
min       5.500000      2.090000
25%       6.100000      4.337500
50%       6.800000      5.010000
75%       7.400000      5.740000
max       8.000000      7.860000
Highest Mean: rainfall_mm
Highest Std Dev: rainfall_mm
```

In [21]:

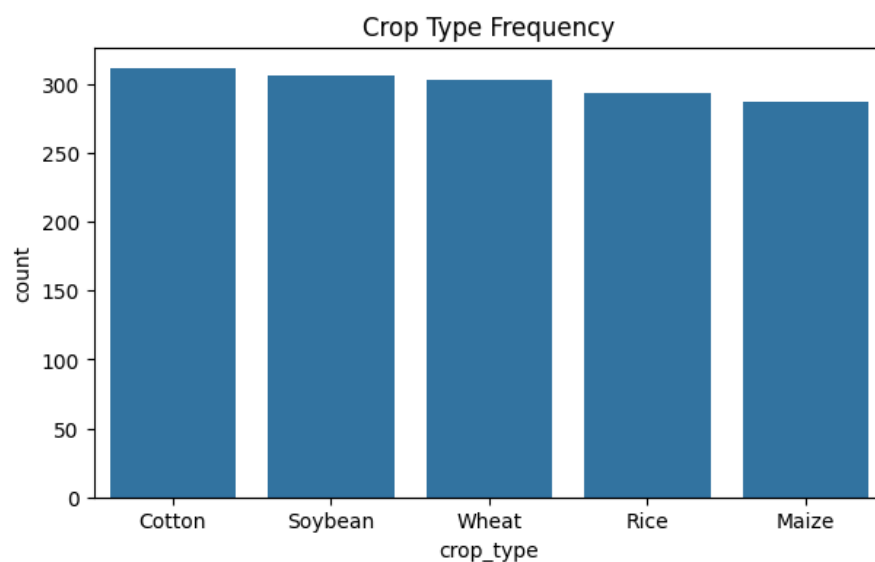
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
fig, axes = plt.subplots(2, 2, figsize=(12, 10))
sns.histplot(df['rainfall_mm'], kde=True, ax=axes[0, 0], color='blue').set_title('Rainfall Distribution')
sns.histplot(df['temperature_c'], kde=True, ax=axes[0, 1], color='orange').set_title('Temperature Distribution')
sns.histplot(df['fertilizer_kg'], kde=True, ax=axes[1, 0], color='green').set_title('Fertilizer Distribution')
sns.histplot(df['yield_ton_per_hectare'], kde=True, ax=axes[1, 1], color='red').set_title('Yield Distribution')
plt.tight_layout()
plt.show()
```



In [22]:

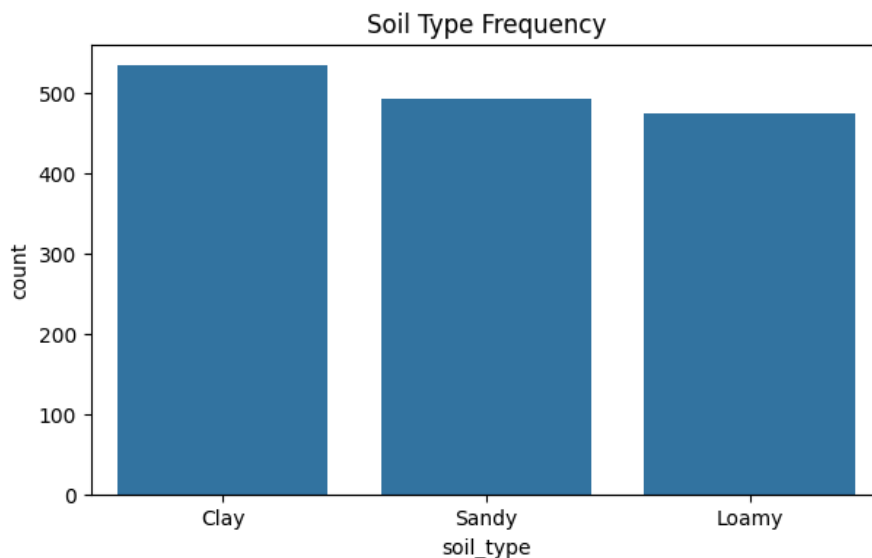
```
print("\nCrop Type Counts:")
print(df['crop_type'].value_counts())
plt.figure(figsize=(7, 4))
sns.countplot(data=df, x='crop_type', order=df['crop_type'].value_counts().index)
plt.title("Crop Type Frequency")
plt.show()
```

```
Crop Type Counts:
crop_type
Cotton      311
Soybean     306
Wheat       303
Rice        293
Maize       287
Name: count, dtype: int64
```

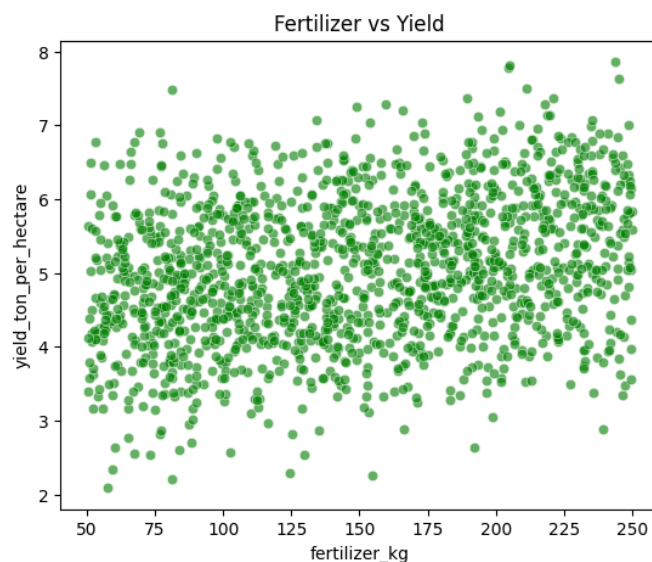
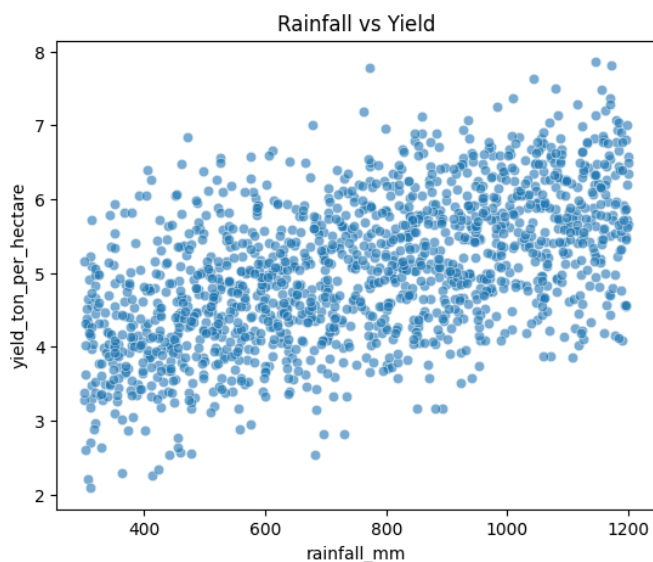


```
In [23]: print("\nSoil Type Counts:")
print(df['soil_type'].value_counts())
plt.figure(figsize=(7, 4))
sns.countplot(data=df, x='soil_type', order=df['soil_type'].value_counts().index)
plt.title("Soil Type Frequency")
plt.show()
```

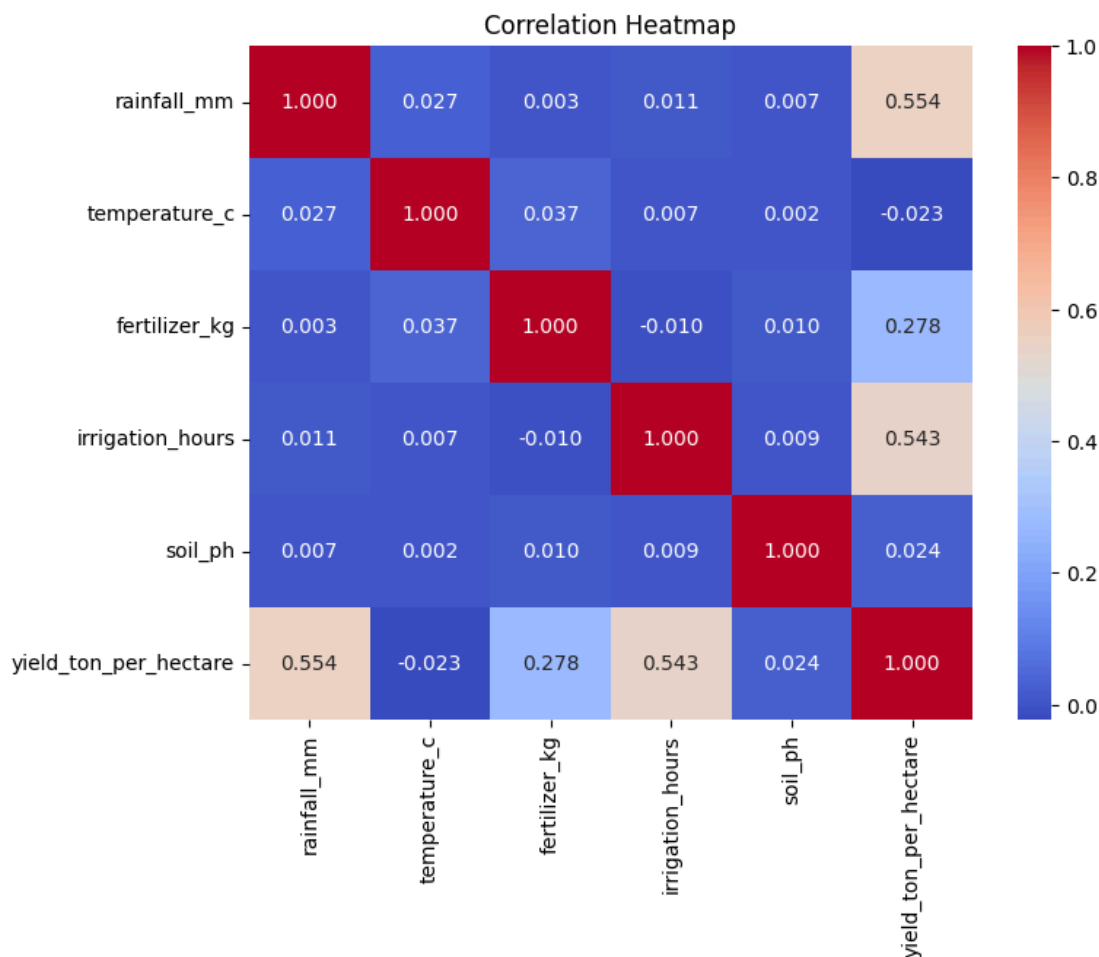
```
Soil Type Counts:
soil_type
Clay      534
Sandy     492
Loamy     474
Name: count, dtype: int64
```



```
In [24]: fig, axes = plt.subplots(1, 2, figsize=(14, 5))
sns.scatterplot(data=df, x='rainfall_mm', y='yield_ton_per_hectare', ax=axes[0], alpha=0.6).set_title('Rainfall vs Yield')
sns.scatterplot(data=df, x='fertilizer_kg', y='yield_ton_per_hectare', ax=axes[1], alpha=0.6, color='g').set_title('Fertilizer vs Yield')
plt.show()
```



```
In [25]: plt.figure(figsize=(8, 6))
numeric_df = df.select_dtypes(include=[np.number])
corr_matrix = numeric_df.corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".3f")
plt.title("Correlation Heatmap")
plt.show()
print("\nCorrelation with yield:")
print(corr_matrix['yield_ton_per_hectare'].sort_values(ascending=False))
```



```
Correlation with yield:
yield_ton_per_hectare    1.000000
rainfall_mm              0.553704
irrigation_hours         0.542664
fertilizer_kg            0.278043
soil_ph                  0.024412
temperature_c            -0.022559
Name: yield_ton_per_hectare, dtype: float64
```

In [26]:

```
print(f"\nMean yield by Crop Type:\n{df.groupby('crop_type')['yield_ton_per_hectare'].mean()}")
print(f"\nMean yield by Soil Type:\n{df.groupby('soil_type')['yield_ton_per_hectare'].mean()}")
```

```
Mean yield by Crop Type:
crop_type
Cotton      4.607299
Maize       4.897143
Rice        5.494744
Soybean     5.173431
Wheat       4.989472
Name: yield_ton_per_hectare, dtype: float64
```

```
Mean yield by Soil Type:
soil_type
Clay        5.134326
Loamy       5.366519
Sandy       4.588882
Name: yield_ton_per_hectare, dtype: float64
```

In [27]:

```
print("\n--- Part C: Data Preparation ---")
df_encoded = pd.get_dummies(df, columns=['crop_type', 'soil_type'], drop_first=True)
print("\nFirst 5 rows of transformed dataset:")
print(df_encoded.head())
```

--- Part C: Data Preparation ---

First 5 rows of transformed dataset:

|   | rainfall_mm | temperature_c | fertilizer_kg | irrigation_hours | soil_ph | \ |
|---|-------------|---------------|---------------|------------------|---------|---|
| 0 | 588.6       | 18.6          | 242.4         | 6.5              | 6.5     |   |
| 1 | 772.8       | 34.6          | 247.2         | 10.0             | 6.5     |   |
| 2 | 970.9       | 36.3          | 168.4         | 7.3              | 6.4     |   |
| 3 | 611.7       | 19.0          | 121.7         | 3.7              | 6.0     |   |
| 4 | 696.1       | 29.6          | 184.6         | 5.1              | 6.1     |   |

|   | yield_ton_per_hectare | crop_type_Maize | crop_type_Rice | crop_type_Soybean | \ |
|---|-----------------------|-----------------|----------------|-------------------|---|
| 0 | 5.92                  | True            | False          | False             |   |
| 1 | 6.24                  | True            | False          | False             |   |
| 2 | 4.77                  | False           | False          | False             |   |
| 3 | 4.04                  | True            | False          | False             |   |
| 4 | 5.17                  | False           | False          | False             |   |

|   |                 |                 |                 |
|---|-----------------|-----------------|-----------------|
|   | crop_type_Wheat | soil_type_Loamy | soil_type_Sandy |
| 0 | False           | True            | False           |
| 1 | False           | False           | True            |
| 2 | True            | False           | True            |
| 3 | False           | True            | False           |
| 4 | False           | False           | False           |

```
In [29]: X = df_encoded.drop(columns=['yield_ton_per_hectare'])
y = df_encoded['yield_ton_per_hectare']
print("\n--- Part D: Machine Learning ---")
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print(f"X_train shape: {X_train.shape}, X_test shape: {X_test.shape}")
print(f"y_train shape: {y_train.shape}, y_test shape: {y_test.shape}")
```

```
--- Part D: Machine Learning ---
X_train shape: (1200, 11), X_test shape: (300, 11)
y_train shape: (1200,), y_test shape: (300,)
```

```
In [30]: model = LinearRegression()
model.fit(X_train, y_train)

print(f"\nModel Intercept: {model.intercept_:.4f}")
coefficients = pd.DataFrame({'Feature': X.columns, 'Coefficient': model.coef_})
print("\nModel Coefficients:")
print(coefficients.sort_values(by='Coefficient', ascending=False))
```

```
Model Intercept: 1.5871
```

```
Model Coefficients:
      Feature  Coefficient
6  crop_type_Rice    0.863993
7  crop_type_Soybean  0.481719
8  crop_type_Wheat   0.348474
9  soil_type_Loamy    0.302974
5  crop_type_Maize    0.238931
3  irrigation_hours  0.198327
2  fertilizer_kg      0.004978
4  soil_ph            0.004901
0  rainfall_mm        0.002035
1  temperature_c     -0.009175
10 soil_type_Sandy   -0.490933
```